

RoHS

safety

APPROVAL SHEET

CUSTOMER : LG Electronics Co., Ltd.

MODEL NO : **LGPS83W6-26OP**
DESCRIPTION : **LLC TRANS**
MAKER NO : **26OP-LM83W**
PART NO : **EBJ30067801**
DATE : **2025. 05. 26.**

REQUEST CUSTOMER	FEELUX ELECTRONICS CO.,LTD (KOREA FACTORY)	TEL	070-7780-8421
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REQUEST CUSTOMER	CHUNGKYE KH FEELUX ELECTRONICS CO.,LTD (SEOUL OFFICE)	TEL	070-7780-8420
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REQUEST CUSTOMER	FEELUX OPTOELECTRONIC (SHANDONG) Co.,Ltd (CHINA FACTORY 1.)	TEL	0086-631-786-7903
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 **KH Feelux 주식회사 필룩스**

RECORD OF REVISION

REV. NO	REASON	CONTENTS	DATE OF APPROVAL	CHECKED	REMARKS
0		신규제정	2025. 05. 26.		

승인원 CHECK LIST (TRANS)

				Checking (MAKER)		Remark
대분류	소분류	내용	CTQ 지정	Yes	No	
Approval	표지	승인원 표지에 P/N 는 정확한가?		☐		
		승인원 표지에 SPEC는 정확한가?		☐		
		생산지 가 정확하게 표기 되었는가?		☐		
		작성일자를 정확히 표기 되었는가?		☐		
		영업/QA담당자명 및 연락처가 정확히 기재되었는가?		☐		
		"Safety"마크는있는가?(우측상단)	★	☐		
	Revision Record	RevisionRecord란의내용은 정확히기재되어있는가?		☐		
		Revision Record 란은 의 Date 는 정확한가?		☐		
		Rev 번호는 정확히 적혀있는가?		☐		
	INDEX	승인원에 포함되어야 할 모든 내용이 list화 되어 있는가?		☐		
	도면	치수 및 공차가 정확히 기재되어 있는가?		☐		
		Marking Spec은 제대로 기재되어 있는가?		☐		
		Bonding의 위치 표기는 제대로 되어 있는가?		☐		
		"Safety" 마크는 있는가?(우측상단)		☐		
		Bobbin의 Pin번호는 표기되어 있는가?		☐		
		Pin간 거리 tolerance는 spec을 만족하는가?		☐		
	Winding Specification	Schematic 도면과 같이 동일한 Pin no로 구성되었는가?		☐		
		권선차수별 올바르게 기재되었는가?		☐		
		Wire권선 종류는 제대로 적용되었는가?		☐		
		권선방법은 제대로 기재되었는가?		☐		
		각 권선별 turn수는 정확한가?	★	☐		
		절연tape의 색상과 size는 올바른가?	★	☐		
	Packing Method	절연tape의 turn수는 정확한가?	★	☐		
		Inner Box의 형상과 제질명기가 정확한가?		☐		
		Inner Box 1PCS당 담을수 있는 샘플 수량이 기재되었는가?		☐		
		Out Box의 형상과 제질명기가 정확한가?		☐		
	UL Card	Out Box 하나당 packing되는 샘플 수량이 정확히 기재되었는가?		☐		
		각 재질군별 난연등급 규격을 취득한 제품으로 적용되었는가?		☐		
	RAW Material List	부품에적용되고있는원자재만기재되어있는가?		☐		
		원자재업체명,부품형명이정확한가?		☐		
	Internal Construction	Trans내부 권선구조가 잘 도식화 되어 있는가?		☐		
	분해검사 성적서	실측치 기입과 측정담당자 명기가 되어 있는가?		☐		
		모든 재질군 분해검사가 실제로 진행되었는가?		☐		
	4M LIST	4M 변경 신고기준서는 첨부되었는가?		☐		
Reliability test	환경시험	테스트 환경조건 명기가 정확한가?		☐		
		사용온도 조건이 명기가 되어있는가?		☐		
		내습성,내열성,내한성 온도/시간 조건이 명확한가?		☐		
		온도Cycle 단계별 조건이 명확한가?		☐		
		열충격테스트조건명기가정확한가?		☐		
		(Cycle단계별적용temp및시간이LGE기준과동일수준인가?)			☐	
	전기적성능	PFOA 재질 포함 되었는가?			☐	
		Inductance값이 각 teminal별로 제대로 기재되어 있는가?		☐		
		Leakage Inductance값이 기재되어 있는가?		☐		
		DC저항값이 기재되어 있는가?		☐		
		내전압 시험조건이 정확히 명기되어 있는가?	★	☐		
		절연저항 값이 제대로 기재되어 있는가?	★	☐		
	기계적성능	Impulse test조건은 제대로 명기되어 있는가?	★	☐		
		충간 short test 기준은 명확한가?		☐		
		Terminal 발거력 spec은 명확한가?		☐		
	4M 관점의 QC공정도	납불임성테스트조건은 명확한가?		☐		
		4M 관점에서 기재되어 있는가?	★	☐		
QC공정도	4M 관점의 QC공정도					
승인원	승인원	승인원의 모든 내용에서 오타가 없는가?		☐		

SPECIFICATION

NO. 4 / 27		LLC TRANS		P / N		26OP-LM83W(EBJ30067801)	
변경사항 CHANGE	(1)			(3)			
	(2)			(4)			
INDEX 4/27 : INDEX 5/27 : 1 DIMENSION 6/27 : 2 SCHEMATIC DIAGRAM : 3 WINDING SPECIFICATION 7/27 : 4 ELECTRICAL CHARACTERISTICS 8/27 : 5 ENVIRONMENTAL CHARACTERISTICS 9/27 : 6 MECHANICAL CHARACTERISTICS 10/27 : 7 PART LIFE 11/27 : 8 INTERNAL CONSTRUCTION 12/27 : 9 BOND APPLY PHOTO 13/27 : 10 MATERIAL LIST 14/27 : 11 UL-LIST 15/27 : UL-LIST 16/27 : UL-LIST 17/27 : 12 CORE DIMENSION 18/27 : 13 BOBBIN DIMENSION 19/27 : 14 PROCESS FLOW 20/27 : 15 PACKING METHOD 21/27 : 16 INSPECTION DATA 22/27 : 17 ASSIGNMENT 23/27 : 18 CHEMICAL ANALYSIS REPORT FOR BOBBIN 24/27 : 19 CHEMICAL ANALYSIS REPORT FOR CP-WIRE 25/27 : 20 WIRE SPECIFICATION 26/27 : 21 QC FLOW CHART 27/27 : 22 4M CATEGORY							
SPEC NO.		DATE		2025. 05. 26.		SHEET NO.	

SPECIFICATION

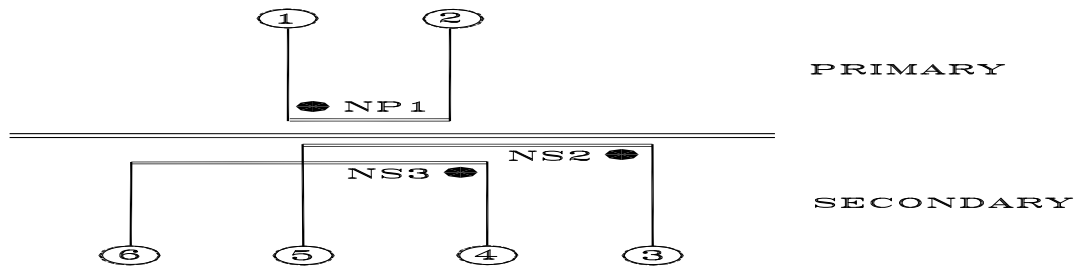
NO. 5 / 27		LLC TRANS		P / N		26OP-LM83W(EBJ30067801)	
변경 사항 CHANGE	(1)		(3)				
	(2)		(4)				

1. DIMENSION

SPECIFICATION

NO. 6 / 27	LLC TRANS	P / N	26OP-LM83W(EBJ30067801)
변경사항 CHANGE	(1)	(3)	
	(2)	(4)	

2. SCHEMATIC DIAGRAM



3. WINDING SPEC

No.	TERMINAL		WIRE	TURNS	WINDING METHOD	WINDING LOCATION		
	S	F						
NP1	1	2	MIW-F(LZ) 0.04C x 360	5 Ts	SOLENOID WINDING	Primary SECTION		
NS2	3	5	Cu plate 0.5T*4.5W	1 Ts		Secondary SECTION		
NS3	4	6	Cu plate 0.5T*4.5W	1 Ts				
CORE FIXING TAPE						P/I TAPE	0.025×21.5 (WHITE)	2Ts MIN

* WINDING DIRECTION : BOTTOM VIEW C.C.W(COUNTER CLOCK WISE)
 * WIRE TOLERANCE : 0.04mm±0.15
 * Requirements for insulating skins after soldering :
 A. Do not exceed the bottom surface of the product (the product itself is bottom surface).
 B. Do not exceed the product support point (sunken product)
 C. Do not melting-loss in the winding

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SPECIFICATION

NO. 7 / 27	LLC TRANS	P / N	26OP-LM83W(EBJ30067801)
변경사항 CHANGE	(1)	(3)	
	(2)	(4)	

4. ELECTRICAL CHARACTERISTICS

NO	ITEM	TERMINAL	SPECIFICATION	REMARKS
4-1	INDUCTANCE	1 -- 2 3 -- 5 4 -- 6	53 uH +7%, -10% 2.80 uH ± 50% 2.80 uH ± 50%	L.C.R METER HP-4284A 3302 AT 100KHz / 1V
4-2	LEAKAGE INDUCTANCE	1 -- 2 (SECONDARY ALL SHORT)	14.0 uH ± 15%	
4-3	D.C RESISTANCE	1 -- 2 3 -- 5 4 -- 6	0.055 Ω MAX 0.010 Ω MAX 0.010 Ω MAX	H10K13224mΩ HI TESTER GOM-802
4-4	WITHSTANDING VOLTAGE	P COIL - S COIL P COIL - CORE	AC <u>3.75</u> KV r.m.s (1 MINUTE) (Mp Condition (AC 4.2KV(RMS) 2Sec) AC <u>3.0</u> KV r.m.s (1 MINUTE) (Mp Condition (AC 3.6KV(RMS) 2Sec)	TOS-8850 KIKUSUI TOS 5302  CUT OFF CURRENT : 5mA
4-5	INSULATION RESISTANCE	P COIL - S COIL	100 M ohm MIN at DC 500V	KIKUSUI TOS 8850 KIKUSUI TOS 5302 
4-6	D.C OVERLAP	1 -- 2	MORE THAN 90% OF INITIAL INDUCTANCE	TEST CURRENT : 5.5 A NORMAL TEMPERATURE
4-7	LAYER SHORT	1 ~ 3 (4-5, 6-2 short)	AC <u>1.5</u> KV Pulse 4 Times	
4-8	QUALTY FACTOR	1 -- 2	150 MIN	L.C.R METER HP-4284A 3302 AT 100KHz / 1.0V

※ When measuring DC RESISTANCE, the measured value is displayed as soon as it is taken while the instrument is ON.

※ After the reliability test, the measured value shall satisfy the initial specification.

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SPECIFICATION

NO. 8 / 27	LLC TRANS	P / N	260P-LM83W(EBJ30067801)
변경 사항 CHANGE	(1)	(3)	
	(2)	(4)	

5. ENVIRONMENTAL CHARACTERISTICS

* STANDARD TEST CONDITION : TEMP 20℃ ± 15℃ HUMIDITY 65% ± 20%

IF WHEN ANY OBJECTION IS SUGGESTED THE TEST CAN BE OPERATED AT 20℃ ± 2℃ 65% ± 5% RH

NO	ARTICLE	SPECIFICATION	TEST CONDITION															
5-1	USED TEMPERATURE RANGE	-30℃ ~ +105℃																
5-2	KEEPING TEMPERATURE	-30℃ ~ +105℃																
5-3	LOW TEMPERATURE RESISTANT	MUST MEET THE ELECTRICAL REQUIREMENTS AND HAS NOT ANY DAMAGE ON THE APPEARANCE AND THE INDUCTANCE VARIATION CAN NOT OVER 10% OF THE INITIAL VALUE 1. Inductance <table><tr><th>Inductance</th><th>Deviation Ratio</th></tr><tr><td>10uH More than</td><td>±10%</td></tr><tr><td>10uH under 1uH</td><td>±10%</td></tr><tr><td>More than 1uH under 0.3uH</td><td>±40%</td></tr><tr><td>More than 0.3uH under</td><td>±0.5uH MAX</td></tr></table>	Inductance	Deviation Ratio	10uH More than	±10%	10uH under 1uH	±10%	More than 1uH under 0.3uH	±40%	More than 0.3uH under	±0.5uH MAX	-25℃ ± 2℃, 96Hr MEASURE AFTER 2 Hr AT ROOM TEMP.					
Inductance	Deviation Ratio																	
10uH More than	±10%																	
10uH under 1uH	±10%																	
More than 1uH under 0.3uH	±40%																	
More than 0.3uH under	±0.5uH MAX																	
5-4	HIGH TEMPERATURE RESISTANT	+105℃ ± 2℃, 96Hr MEASURE AFTER 2 Hr AT ROOM TEMP.																
5-5	HUMIDITY RESISTANT	+95%RH ± 5% (60℃ ± 5℃) 200Hr MEASURE AFTER 2Hr AT ROOM TEMP																
5-6	TEMPERATURE CYCLE	2. Leakage - Can not over 10% of the initial value. 3. DCR - Can not over MAX of the SPEC. 4. DC overlap - It should be over 90% in Indutance 5. Q - Can not over MIN of the SPEC	<table><tr><th>STEP</th><th>TEMP</th><th>TIME</th></tr><tr><td>1</td><td>-45℃ ± 2℃</td><td>30 MINUTE</td></tr><tr><td>2</td><td>+25℃ ± 2℃</td><td>5 MINUTE</td></tr><tr><td>3</td><td>+105℃ ± 2℃</td><td>30 MINUTE</td></tr><tr><td>4</td><td>+25℃ ± 2℃</td><td>5 MINUTE</td></tr></table> REPEAT 10 TIME FROM STEP 1 AND TEST AFTER 2 Hr. AT ROOM TEMP.	STEP	TEMP	TIME	1	-45℃ ± 2℃	30 MINUTE	2	+25℃ ± 2℃	5 MINUTE	3	+105℃ ± 2℃	30 MINUTE	4	+25℃ ± 2℃	5 MINUTE
STEP	TEMP	TIME																
1	-45℃ ± 2℃	30 MINUTE																
2	+25℃ ± 2℃	5 MINUTE																
3	+105℃ ± 2℃	30 MINUTE																
4	+25℃ ± 2℃	5 MINUTE																

※ After the reliability test, the measured value shall satisfy the initial specification.

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SPECIFICATION

NO. 9 / 27	LLC TRANS	P / N	260P-LM83W(EBJ30067801)
변경사항 CHANGE	(1)	(3)	
	(2)	(4)	

6. MECHANICAL CHARACTERISTICS

NO	ARTICLE	SPECIFICATION	TEST CONDITION															
6-1	LAYER SHORT TEST	THERE IS NO DEFECT ON THE WIRE INSULATION COMPARE WITH STANDARD SAMPLE	SUPPLY SURGE VOLTAGE (AT 1.5KV 60Hz for 3SECOND) BETWEEN WIRE AND MEASURE THE WIRE INSULATION															
6-2	SOLDER HEAT TEST	MUST MEET THE ELECTRICAL REQUIREMENTS AND HAS NOT ANY DAMAGE ON THE APPEARANCE AND THE INDUCTANCE VARIATION CAN NOT OVER 10% OF THE INITIAL VALUE 1. Indutance - It should be managed by $\pm 10\%$ from more than $10\mu H$ - It should be managed by $\pm 10\%$ from $10\mu H$ TO $1\mu H$ - It should be managed by $\pm 40\%$ from under $1\mu H$ to $0.3\mu H$ - It should be managed by $0.5\mu H$ max. under $0.3\mu H$ 2. Leakage - Can not over 10% of the initial value. 3. DCR - Can not over MAX of the SPEC. 4. DC overlap - It should be over 90% in Indutance 5. Q - Can not over MIN of the SPEC	DIP THE TERMINAL IN SOLDER POT AT $260^{\circ}C$ $-0/+5^{\circ}C$ FOR 10 SECONDS MEASURE AFTER 2Hr AT ROOM TEMP.															
6-3	SOLDER ABILITY	95% MIN OF THE PIN OUT-DIA BE COVERED BY NEW SOLDER	DIP THE TERMINAL IN SOLDER AT $245^{\circ}C$ $-5^{\circ}C$ FOR 3 SECONDS															
6-4	TERMINAL PULL STRENGTH	THE PRODUCT HAS NOT ANY ABNORMALITY AND OPERATE PROPERLY SATISFIED INITIAL VALUE	<table><tr><th>공칭단면적(mm²)</th><th>공칭선경(mm)</th><th>인장력N (Kgf)</th></tr><tr><td>0.07<S≤0.2</td><td>0.3<d≤0.5</td><td>5(0.5)</td></tr><tr><td>0.2<S≤0.5</td><td>0.5<d≤0.8</td><td>10(1.0)</td></tr><tr><td>0.5<S≤1.0</td><td>0.8<d≤1.2</td><td>20(2.0)</td></tr><tr><td>1.0<S</td><td>1.2<d</td><td>45(4.5)</td></tr></table> DURING 30 ± 5 SEC	공칭단면적(mm ²)	공칭선경(mm)	인장력N (Kgf)	0.07<S≤0.2	0.3<d≤0.5	5(0.5)	0.2<S≤0.5	0.5<d≤0.8	10(1.0)	0.5<S≤1.0	0.8<d≤1.2	20(2.0)	1.0<S	1.2<d	45(4.5)
공칭단면적(mm ²)	공칭선경(mm)	인장력N (Kgf)																
0.07<S≤0.2	0.3<d≤0.5	5(0.5)																
0.2<S≤0.5	0.5<d≤0.8	10(1.0)																
0.5<S≤1.0	0.8<d≤1.2	20(2.0)																
1.0<S	1.2<d	45(4.5)																

※ After the reliability test, the measured value shall satisfy the initial specification.

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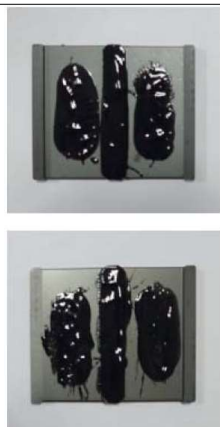
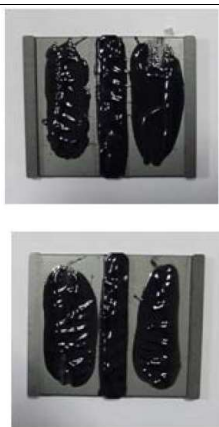
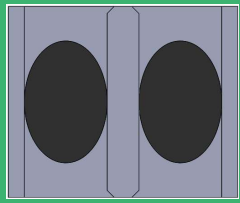
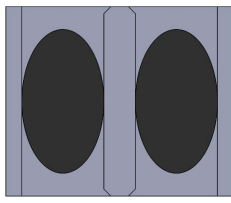
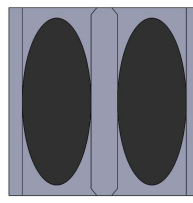
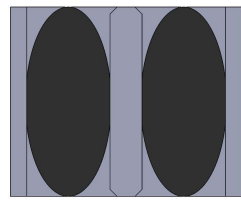
SPECIFICATION

NO.	10 / 27	LLC TRANS	P / N	260P-LM83W(EBJ30067801)																								
변경 사항 CHANGE	(1)		(3)																									
	(2)		(4)																									
7. PART LIFE																												
LIFE TIME	TEST CONDITION																											
B1/TTT LIFE : It must be at least 60,000 hours.	. 가속인자 : 온도																											
	. 가속모델 : 아레니우스 모형 (Arrhenius model)																											
	. 활성화에너지 : 0.7eV 적용																											
	. 사용온도 : 부품 온도 Max Spec 80% 적용 69℃ (분위기 온도 25℃ 기준 / TV제품온도(CST) 40℃)																											
	아레니우스모형(Arrhenius model) AF = exp[(E/k)·(1/T_d - 1/T_a)]																											
	가속인자:온도																											
	가속계수 공식																											
	$A_T = \text{EXP} \left\{ \frac{E_a}{k} \left(\frac{1}{273 + T_{\text{field}}} - \frac{1}{273 + T_{\text{test}}} \right) \right\}$																											
	E_a = Activation energy (활성화에너지, eV) → 물질에 따라 달라짐 화학 반응이 진행되기 위해 필요한 최소한의 에너지를 의미함 k = Boltzmann constant = 8.617 x 10⁻⁵ eV/K T_{test} = Test temperature (℃) → 공식에는 절대온도로 적용함 T_{field} = Field temperature (℃) → 공식에는 절대온도로 적용함																											
	. B1 수명 계산식																											
<table><tr><td>Stress/Operation Condition</td><td>Temperature</td><td>STRESS CONDITION (Test Condition)</td><td>125</td><td>℃</td><td></td></tr><tr><td></td><td></td><td>OPERATION CONDITION</td><td>69</td><td>℃</td><td></td></tr></table>		Stress/Operation Condition	Temperature	STRESS CONDITION (Test Condition)	125	℃				OPERATION CONDITION	69	℃		<table><tr><td>Acceleration Factor</td><td>Temperature</td><td>Acceleration Factor</td><td>28.28</td><td></td><td>아레니우스</td></tr><tr><td></td><td></td><td>Parameter (활성화 에너지)</td><td>0.7</td><td>eV</td><td></td></tr></table>			Acceleration Factor	Temperature	Acceleration Factor	28.28		아레니우스			Parameter (활성화 에너지)	0.7	eV	
Stress/Operation Condition	Temperature	STRESS CONDITION (Test Condition)	125	℃																								
		OPERATION CONDITION	69	℃																								
Acceleration Factor	Temperature	Acceleration Factor	28.28		아레니우스																							
		Parameter (활성화 에너지)	0.7	eV																								
Sample Size		90	ea																									
Test Time		2160	Hours																									
Confidence Level		60	%																									
MTTF		6,007,692	Hours																									
B1 life		60,379	Hours																									
. TTT 수명 계산식																												
<table><tr><td>Stress/Operation Condition</td><td>Temperature</td><td>STRESS CONDITION (Test Condition)</td><td>125</td><td>℃</td><td></td></tr><tr><td></td><td></td><td>OPERATION CONDITION</td><td>69</td><td>℃</td><td></td></tr></table>		Stress/Operation Condition	Temperature	STRESS CONDITION (Test Condition)	125	℃				OPERATION CONDITION	69	℃		<table><tr><td>Acceleration Factor</td><td>Temperature</td><td>Acceleration Factor</td><td>28.28</td><td></td><td>아레니우스</td></tr><tr><td></td><td></td><td>Parameter (활성화 에너지)</td><td>0.7</td><td>eV</td><td></td></tr></table>			Acceleration Factor	Temperature	Acceleration Factor	28.28		아레니우스			Parameter (활성화 에너지)	0.7	eV	
Stress/Operation Condition	Temperature	STRESS CONDITION (Test Condition)	125	℃																								
		OPERATION CONDITION	69	℃																								
Acceleration Factor	Temperature	Acceleration Factor	28.28		아레니우스																							
		Parameter (활성화 에너지)	0.7	eV																								
Sample Size		10	ea																									
Test Time		2160	Hours																									
Confidence Level		60	%																									
TTT 수명		61,078	Hours	TTT수명=가속계수*시험시간																								
SPEC NO.		DATE	2025. 05. 26.	SHEET NO.																								

SPECIFICATION

NO. 11 / 27		LLC TRANS		P / N		26OP-LM83W(EBJ30067801)	
변경사항 CHANGE	(1)		(3)				
	(2)		(4)				
8. INTERNAL CONSTRUCTION ⑤ NS4 ③ NS3 ③ ③ NP1 ① GAP CORE ③ NP1 ③ NS4 ③ NS3 ③ ⑤ ② ① NO GAP CORE ② TOP BOTTOM ⑥ CENTER BONDING 100% APPLY ⑤ NS4 ③ NS3 ③ ③ NP1 ① GAP CORE ③ NP1 ③ NS4 ③ NS3 ③ ⑤ ② ① NO GAP CORE ② TOP BOTTOM ⑥ CENTER BONDING 100% APPLY ⑤ NS4 ③ NS3 ③ ③ NP1 ① GAP CORE ③ NP1 ③ NS4 ③ NS3 ③ ⑤ ② ① NO GAP CORE ② TOP BOTTOM ⑥ CENTER BONDING 100% APPLY ⑤ NS4 ③ NS3 ③ ③ NP1 ① GAP CORE ③ NP1 ③ NS4 ③ NS3 ③ ⑤ ② ① NO GAP CORE ② TOP BOTTOM ⑥ CENTER BONDING 100% APPLY ⑤ NS4 ③ NS3 ③ ③ NP1 ① GAP CORE ③ NP1 ③ NS4 ③ NS3 ③ ⑤ ② ① NO GAP CORE ② TOP BOTTOM ⑥ CENTER BONDING 100% APPLY ⑤ NS4 ③ NS3 ③ ③ NP1 ① GAP CORE ③ NP1 ③ NS4 ③ NS3 ③ ⑤ ② ① NO GAP CORE ② TOP BOTTOM ⑥ CENTER BONDING 100% APPLY ⑤ NS4 ③ NS3 ③ ③ NP1 ① GAP CORE ③ NP1 ③ NS4 ③ NS3 ③ ⑤ ② ① NO GAP CORE ② TOP BOTTOM ⑥ CENTER BONDING 100% APPLY ⑤ NS4 ③ NS3 ③ ③ NP1 ① GAP CORE ③ NP1 ③ NS4 ③ NS3 ③ ⑤ ② ① NO GAP CORE ② TOP BOTTOM ⑥ CENTER BONDING 100% APPLY ⑤ NS4 ③ NS3 ③ ③ NP1 ① GAP CORE ③ NP1 ③ NS4 ③ NS3 ③ ⑤ ② ① NO GAP CORE ② TOP BOTTOM ⑥ CENTER BONDING 100% APPLY ⑤ NS4 ③ NS3 ③ ③ NP1 ① GAP CORE ③ NP1 ③ NS4 ③ NS3 ③ ⑤ ② ① NO GAP CORE ② TOP BOTTOM ⑥ CENTER BONDING 100% APPLY ⑤ NS4 ③ NS3 ③ ③ NP1 ① GAP CORE ③ NP1 ③ NS4 ③ NS3 ③ ⑤ ② ① NO GAP CORE ② TOP BOTTOM ⑥ CENTER BONDING 100% APPLY ⑤ NS4 ③ NS3 ③ ③ NP1 ① GAP CORE ③ NP1 ③ NS4 ③ NS3 ③ ⑤ ② ① NO GAP CORE ② TOP BOTTOM ⑥ CENTER BONDING 100% APPLY ⑤ NS4 ③ NS3 ③ ③ NP1 ① GAP CORE ③ NP1 ③ NS4 ③ NS3 ③ ⑤ ② ① NO GAP CORE ② TOP BOTTOM ⑥ CENTER BONDING 100% APPLY ⑤ NS4 ③ NS3 ③ ③ NP1 ① GAP CORE ③ NP1 ③ NS4 ③ NS3 ③ ⑤ ② ① NO GAP CORE ② TOP BOTTOM ⑥ CENTER BONDING 100% APPLY ⑤ NS4 ③ NS3 ③ ③ NP1 ① GAP CORE ③ NP1 ③ NS4 ③ NS3 ③ ⑤ ② ① NO GAP CORE ② TOP BOTTOM ⑥ CENTER BONDING 100% APPLY ⑤ NS4 ③ NS3 ③ ③ NP1 ① GAP CORE ③ NP1 ③ NS4 ③ NS3 ③ ⑤ ② ① NO GAP CORE ② TOP BOTTOM ⑥ CENTER BONDING 100% APPLY ⑤ NS4 ③ NS3 ③ ③ NP1 ① GAP CORE ③ NP1 ③ NS4 ③ NS3 ③ ⑤ ② ① NO GAP CORE ② TOP BOTTOM ⑥ CENTER BONDING 100% APPLY ⑤ NS4 ③ NS3 ③ ③ NP1 ① GAP CORE ③ NP1 ③ NS4 ③ NS3 ③ ⑤ ② ① NO GAP CORE ② TOP BOTTOM ⑥ CENTER BONDING 100% APPLY ⑤ NS4 ③ NS3 ③ ③ NP1 ① GAP CORE ③ NP1 ③ NS4 ③ NS3 ③ ⑤ ② ① NO GAP CORE ② TOP BOTTOM ⑥ CENTER BONDING 100% APPLY ⑤ NS4 ③ NS3 ③ ③ NP1 ① GAP CORE ③ NP1 ③ NS4 ③ NS3 ③ ⑤ ② ① NO GAP CORE ② TOP BOTTOM ⑥ CENTER BONDING 100% APPLY ⑤ NS4 ③ NS3 ③ ③ NP1 ① GAP CORE ③ NP1 ③ NS4 ③ NS3 ③ ⑤ ② ① NO GAP CORE ② TOP BOTTOM ⑥ CENTER BONDING 100% APPLY ⑤ NS4 ③ NS3 ③ ③ NP1 ① GAP CORE ③ NP1 ③ NS4 ③ NS3 ③ ⑤ ② ① NO GAP CORE ② TOP BOTTOM ⑥ CENTER BONDING 100% APPLY ⑤ NS4 ③ NS3 ③ ③ NP1 ① GAP CORE ③ NP1 ③ NS4 ③ NS3 ③ ⑤ ② ① NO GAP CORE ② TOP BOTTOM ⑥ CENTER BONDING 100% APPLY ⑤ NS4 ③ NS3 ③ ③ NP1 ① GAP CORE ③ NP1 ③ NS4 ③ NS3 ③ ⑤ ② ① NO GAP CORE ② TOP BOTTOM ⑥ CENTER BONDING 100% APPLY ⑤ NS4 ③ NS3 ③ ③ NP1 ① GAP CORE ③ NP1 ③ NS4 ③ NS3 ③ ⑤ ② ① NO GAP CORE ② TOP BOTTOM ⑥ CENTER BONDING 100% APPLY ⑤ NS4 ③ NS3 ③ ③ NP1 ① GAP CORE ③ NP1 ③ NS4 ③ NS3 ③ ⑤ ② ① NO GAP CORE ② TOP BOTTOM ⑥ CENTER BONDING 100% APPLY ⑤ NS4 ③ NS3 ③ ③ NP1 ① GAP CORE ③ NP1 ③ NS4 ③ NS3 ③ ⑤ ② ① NO GAP CORE ② TOP BOTTOM ⑥ CENTER BONDING 100% APPLY ⑤ NS4 ③ NS3 ③ ③ NP1 ① GAP CORE ③ NP1 ③ NS4 ③ NS3 ③ ⑤ ② ① NO GAP CORE ② TOP BOTTOM ⑥ CENTER BONDING 100% APPLY ⑤ NS4 ③ NS3 ③ ③ NP1 ① GAP CORE ③ NP1 ③ NS4 ③ NS3 ③ ⑤ ② ① NO GAP CORE ② TOP BOTTOM ⑥ CENTER BONDING 100% APPLY ⑤ NS4 ③ NS3 ③ ③ NP1 ① GAP CORE ③ NP1 ③ NS4 ③ NS3 ③ ⑤ ② ① NO GAP CORE ② TOP BOTTOM ⑥ CENTER BONDING 100% APPLY ⑤ NS4 ③ NS3 ③ ③ NP1 ① GAP CORE ③ NP1 ③ NS4 ③ NS3 ③ ⑤ ② ① NO GAP CORE ② TOP BOTTOM ⑥ CENTER BONDING 100% APPLY ⑤ NS4 ③ NS3 ③ ③ NP1 ① GAP CORE ③ NP1 ③ NS4 ③ NS3 ③ ⑤ ② ① NO GAP CORE ② TOP BOTTOM ⑥ CENTER BONDING 100% APPLY ⑤ NS4 ③ NS3 ③ ③ NP1 ① GAP CORE ③ NP1 ③ NS4 ③ NS3 ③ ⑤ ② ① NO GAP CORE ② TOP BOTTOM ⑥ CENTER BONDING 100% APPLY ⑤ NS4 ③ NS3 ③ ③ NP1 ① GAP CORE ③ NP1 ③ NS4 ③ NS3 ③ ⑤ ② ① NO GAP CORE ② TOP BOTTOM ⑥ CENTER BONDING 100% APPLY ⑤ NS4 ③ NS3 ③ ③ NP1 ① GAP CORE ③ NP1 ③ NS4 ③ NS3 ③ ⑤ ② ① NO GAP CORE ② TOP BOTTOM ⑥ CENTER BONDING 100% APPLY ⑤ NS4 ③ NS3 ③ ③ NP1 ① GAP CORE ③ NP1 ③ NS4 ③ NS3 ③ ⑤ ② ① NO GAP CORE ② TOP BOTTOM ⑥ CENTER BONDING 100% APPLY ⑤ NS4 ③ NS3 ③ ③ 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③ NS4 ③ NS3 ③ ⑤ ② ① NO GAP CORE ② TOP BOTTOM ⑥ CENTER BONDING 100% APPLY ⑤ NS4 ③ NS3 ③ ③ NP1 ① GAP CORE ③ NP1 ③ NS4 ③ NS3 ③ ⑤ ② ① NO GAP CORE ② TOP BOTTOM ⑥ CENTER BONDING 100% APPLY ⑤ NS4 ③ NS3 ③ ③ NP1 ① GAP CORE ③ NP1 ③ NS4 ③ NS3 ③ ⑤ ② ① NO GAP CORE ② TOP BOTTOM ⑥ CENTER BONDING 100% APPLY ⑤ NS4 ③ NS3 ③ ③ NP1 ① GAP CORE ③ NP1 ③ NS4 ③ NS3 ③ ⑤ ② ① NO GAP CORE ② TOP BOTTOM ⑥ CENTER BONDING 100% APPLY ⑤ NS4 ③ NS3 ③ ③ NP1 ① GAP CORE ③ NP1 ③ NS4 ③ NS3 ③ ⑤ ② ① NO GAP CORE <							

SPECIFICATION

NO.	12 / 27	LLC TRANS	P / N	26OP-LM83W(EBJ30067801)
변경 사항 CHANGE	(1)		(3)	
	(2)		(4)	
※ Center Bonding: 0.50g~0.65g ※ Side Bonding () : 0.20(Min)~0.40g(Max) Side Bonding Apply Point : 8Point Side Bonding Apply Area : 3216mm² (Apply more than 50% of the total area) Side Bonding Type Name : SW-860HF ※ Bonding Machine : ESD SAFE or Air Tac ※ It is guranteed that more than 50% of the Side Bond application area is applie				
9. BOND APPLY PHOTO				
CENTER & SIDE BOND				
MIN		MAX		
				
Picture of side bond application area				
50% Area	60% Area	70% Area	80% Area	
				
SPEC NO.		DATE	2025. 05. 26.	SHEET NO.

SPECIFICATION

NO. 13 / 27		LLC TRANS		P / N		26OP-LM83W(EBJ30067801)	
변경 사항 CHANGE	(1)			(3)			
	(2)			(4)			
10. MATERIAL LIST.							
safety							
NO.	PART	MATERIAL NAME	UL FILE NO	MANUFACTURES		MATERIAL COLOR	
1	CORE	FERRITE EE6405: GP98		NANTONG GUANYOUDA MAGNETIC.,LTD		BLACK	
2	BOBBIN	94V-0 PHENOL RESIN EE6405 V 2P SMD LCP E4008, E4008 B(j)(f2)	E249884	SUMITOMO SUMITOMO CHEMICAL CO LTD		BLACK	
			BOBBIN MANUFACTURER : MERIGO, JIARUI				
		TERMINAL	JIANGYIN LIUHUAN(SIX-CIRQUE) ALLOY WIRE CO LTD		SILVER		
3	WIRE	HY-MIW-F(LZ)@ 0.04Φ × 360	E470559	Huizhou Huaying Electronic Technology Co Ltd		IVORY	
		Copper Plate 0.5T * 4.5W		SHEN ZHEN ZI GAO METAL MATELALS., LTD.		SILVER	
4	CORE FIXING TAPE	Polyimide FILM TAPE CAT NO . PF* 0.025t × 21.5mm	E165111	JINGJIANG YAHUA PRESSURE SENSITIVE GLUE CO LTD		AMBER	
5	INSULATION TAPE	Polyimide FILM TAPE CAT NO . PF* 0.025t × 38.0mm 2SHEET	E165111	JINGJIANG YAHUA PRESSURE SENSITIVE GLUE CO LTD		AMBER	
5	CENTER BOND	SW-860HF		SUNG WOO ES CO., LTD		BLACK	
6	SIDE BOND	SW-860HF		SUNG WOO ES CO., LTD		BLACK	
7	SOLDER	HSE-09		LT Materials Co., LTD		SILVER	
		SC07		TIANJIN QINGHE SCIENCE-TECH DEVELOPMENT CO.,LTD			
8	FLUX	8EB34		KAIFENG SONGQING ELECTRONIC MATERIALS CO., LTD		LEMON YELLOW	
9	MARKING	INK & LABEL		BONMARQUE		BLACK	
				VIDEOJET TECHNOLOGIES(SHANGHAI)CO LTD			
				LAZER MARKING			
SPEC NO.		DATE		2025. 05. 26.		SHEET NO.	

SPECIFICATION

NO. 14 / 27		LLC TRANS		P / N		26OP-LM83W(EBJ30067801)		
변경사항 CHANGE	(1)			(3)				
	(2)			(4)				
11. UL CARD								
* YELLOW CARD								
INSULATION SYSTEM	KH Feelux Co., Ltd. E302847							
	235-48 Gwangjeok-ro							
	Gwangjeok-myeon							
	Yangju-si, Gyeonggi-do 482-844 Republic of Korea							
	Marking: Company name or tradename "FL" , and model designation.							
Note: For additional marking information, refer to the Guide Information Page.								
Class 130(B) Electrical Insulation Systems, System Designation(s): B, B01, B02								
Last Updated on 2025-08-05								
1.BOBBIN	Component - Plastics E249884							
	SUMITOMO CHEMICAL CO LTD							
	ADVANCED MATERIALS DIV, Tokyo Nihombashi Tower, 2-7-1 Nihonbashi, Chuo-Ku Tokyo 103-6020 JP							
	E4008. E4008 B(i)(f2)							
	Liquid Crystal Polymer (LCP) "SUMIKASUPER", furnished as pellets							
	Color	Min. Thk (mm)	Flame Class	HWI	HAI	RTI Elec (°C)	RTI Imp (°C)	RTI Str (°C)
	NC, BK	0.3V-0		3	0	240	200	240
		0.75V-0		3	0	240	220	240
		1.5V-0		2	0	240	220	240
		3V-0		1	0	240	220	240
Comparative Tracking Index (CTI): 3 Inclined Plane Tracking (IPT) kV: -								
Dielectric Strength (kV/mm): - Volume Resistivity (10 ⁹ ohm-cm): -								
High-Voltage Arc Tracking Rate (HVTR): 0 Surface Resistivity (10 ⁹ ohms/square): -								
Dimensional Change (%): - High Volt, Low Current Arc Resis (D495): 5								
(f2) - Subjected to one or more of the following tests: Ultraviolet Light, Water Exposure or Immersion in accordance with UL 746C, where the acceptability for outdoor use is to be determined by UL.								
(i) - Virgin and Regrind up to 50% by weight inclusive have the same basic material characteristics with regard to V-0, HWI, HAI, CTI, Tensile Strength, HDT and RTI only.								
ANSI/UL 94 small-scale test data does not pertain to building materials, furnishings and related contents. ANSI/UL 94 small-scale test data is intended solely for determining the flammability of plastic materials used in the components and parts of end-product devices and appliances, where the acceptability of the combination is determined by UL.								
SPEC NO.		DATE		2025. 05. 26.		SHEET NO.		

SPECIFICATION

NO.	15 / 27	LLC TRANS	P / N	26OP-LM83W(EBJ30067801)
변경사항 CHANGE	(1)		(3)	
	(2)		(4)	
* YELLOW CARD				
2. WIRE	Single- and Multi- layer Insulated Winding Wire			
	E470559			
	Huizhou Huaying Electronic Technology Co Ltd			
	Xikeng Industry Park, Zhongkai High-tech Zone, Huizhou Guangdong 516229 CN			
	HY-MIW-F(LZ)			
	Insulation	Size Range	Temp Class	Rated Volts
	Type	TIW Conductor	(°C)	UL 60950-1 Vpeak
				UL 60601-1 Vrms
				UL 61800-5-1 Vrms
	Reinforced	0.0079-4.96 mm2	155(F)	-
Last Revised:		2024-07-24	© 2026 UL Solutions	
SPEC NO.		DATE	2025. 05. 26.	SHEET NO.

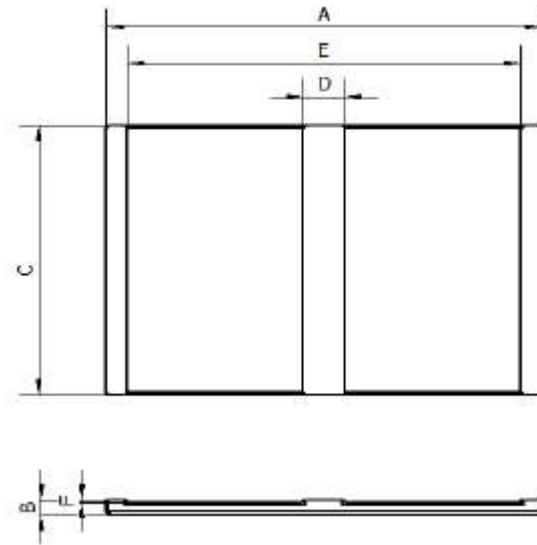
SPECIFICATION

NO. 16 / 27		LLC TRANS		P / N		26OP-LM83W(EBJ30067801)	
변경 사항 CHANGE	(1)		(3)				
	(2)		(4)				
* YELLOW CARD							
3. TAPE	JINGJIANG YAHUA PRESSURE SENSITIVE GLUE CO LTD SOUTHWEST AROUND RD JINGJIANG, JIANGSU 214500 China						E165111
	Cat. No.	Color	Temp (°C)	Flame Retardant+	Optional Ratings Sunlight Resistant+	Cold Resistant+	
	Nonwoven cloth/polyethylene terephthalate film insulating tapes						
	WF* (c)(h)	White	130	Yes	No	No	
	PET film with Acrylic adhesive						
	CT-285B	Black	130	No	No	Yes	
	PET film with polyester non-woven fabric reinforcement and rubber adhesive tape						
	WFF(f)	Brown	130	No	No	No	
	PET films with acrylic adhesive insulating tape						
	CT-280B	Yellow	130	No	No	Yes	
	CT-282F	Blue	130	Yes	No	Yes	
	CT-286F	White	130	Yes	No	Yes	
	Polyethylene terephthalate film insulating tapes						
	CT (b)(g)	Clear	130	No	No	No	
	CT* (b)(g)	All except Clear	130	Yes	No	No	
	CT* (c)(g)	Yellow	130	Yes	No	No	
	PZ* (b)	Yellow	130	Yes	No	No	
	Polyimide film insulating tapes						
	PF* (d)(g)	Amber	180	Yes	No	No	
	+ Complies with "Flame Retardant" and/or "Cold Resistant"and/or "Sunlight Resistant" requirements if authorized in the table above and when so marked. (b) - Comparative Tracking Index (CTI) performance indicates material Group IIIa, PLC=2, CTI equal to or greater than 250 but less than 400 v. (c) - Comparative Tracking Index (CTI) performance indicates material Group I, PLC=0, CTI equal to or greater than 600 v. (d) - Comparative Tracking Index (CTI) performance indicates material Group IIIb, PLC=4, CTI equal to or greater than 100 v but less than 175v. (f) - The CTI test was conducted indicates material Group I, PLC=0, CTI equal to or greater than 600 v per IEC 60112, 4th Edition 2003 and the assigned level is based on the testing of both film and adhesive sides. (g) - The CTI test was conducted per IEC 60112, 4th Edition 2003 and the assigned level is based on the testing of both film and adhesive sides. (h) - The CTI test was conducted per IEC 60112, 4th Edition 2003 and the assigned level is based on the testing of both film and adhesive sides. * - May be followed by suffixes						
Marking: Company name or tradename "E165111", catalog designation printed On the central paper core Or outer package.							
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SPECIFICATION

NO.	17 / 27	LLC TRANS	P / N	260P-LM83W(EBJ30067801)
변경사항 CHANGE	(1)		(3)	
	(2)		(4)	

12.CORE DIMENSION



TYPE	DIMENSIONS IN mm							Unit[mm]
	A	B	C	D	E	F	B-F	
EE6405	64.00±0.80	2.30±0.10	33.50±0.50	8.0±0.25	56.00±0.80	0.70±0.10	1.6±0.15	GUANYOUDA

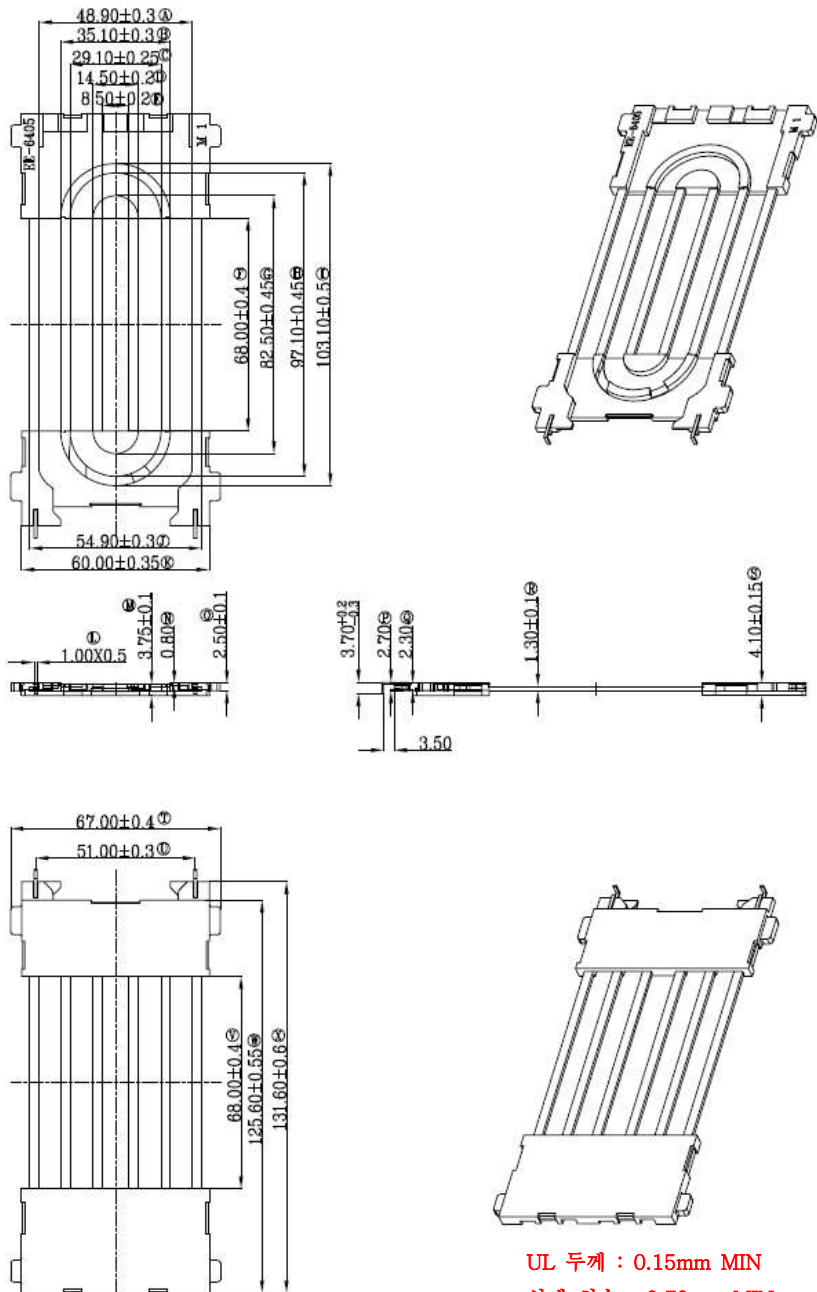
TYPE	EFFECTIVE PRARAMETERS			Unit[mm]
	Le(mm)	Ae(mm²)	Ve(mm³)	
EE6405	57.13	113.1	6461	GUANYOUDA

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SPECIFICATION

NO.	18 / 27	LLC TRANS	P / N	260P-LM83W(EBJ30067801)
변경 사항 CHANGE	(1)		(3)	
	(2)		(4)	

13.BOBBIN DIMENSION



UL 두께 : 0.15mm MIN
설계 치수 : 0.70mm MIN

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KH Feelux 주식회사 KH 필룩스

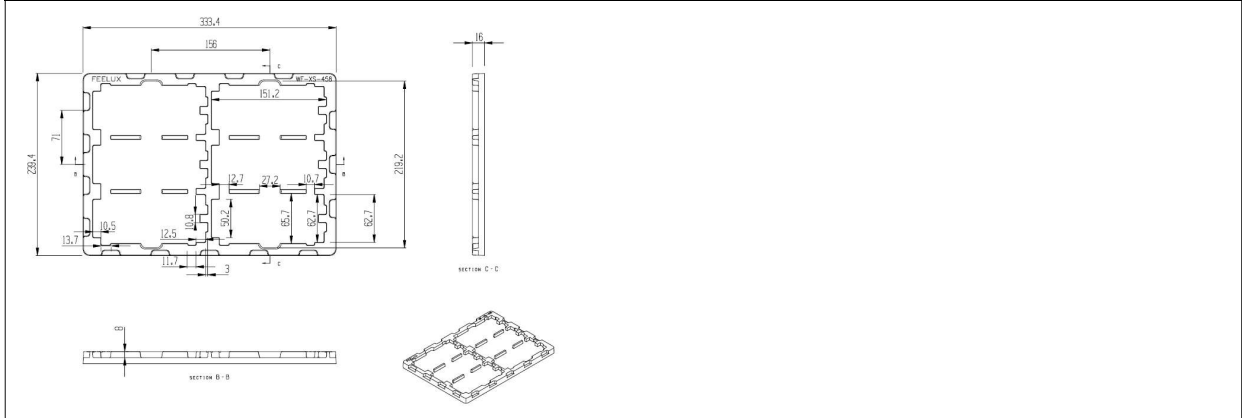
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SPECIFICATION

NO. 20 / 27		LLC TRANS		P / N		26OP-LM83W(EBJ30067801)	
변경사항 CHANGE	(1)			(3)			
	(2)			(4)			

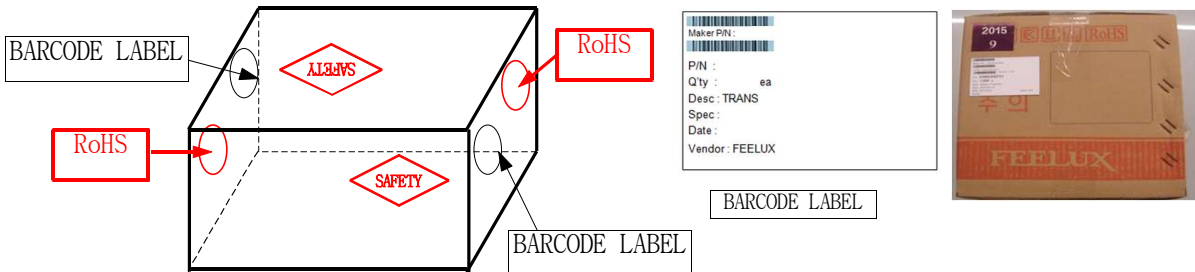
15. PACKING METHOD

1. IN BOX (3 PCS x 2 PCS = 6 PCS)



*IN BOX =TRAY

2. OUT BOX (6 PCS x 15 PCS = 90 PCS))



* In the packaging process, the tray should be zigzag

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SPECIFICATION

NO. 21 / 27		LLC TRANS			P / N		260P-LM83W(EBJ30067801)					
변경사항 CHANGE	(1)			(3)								
	(2)			(4)								
16.INSPECTION DATA												
ITEM NAME		LLC TRANS					TEST DATE		2025. 05. 26.			
PART NO.		260P-LM83W					INSTRUMENT		HP-4284A			
항목	Terminal	Limit	#1	#2	#3	#4	#5	#6	#7	#8	#9	#10
L(uH) (100kHz 1.0v)	1 -- 2	53.0 uH +7%, -10%										
	3 -- 5	2.80 uH ± 30%										
	4 -- 6	2.80 uH ± 30%										
LEAKAGE INDUCTANCE	1 -- 2	14.0 uH ± 15%										
	(SECONDARY ALL SHORT)											
DCR(Ω)	1 -- 2	0.055 Ω MAX										
	3 -- 5	0.010 Ω MAX										
	4 -- 6	0.010 Ω MAX										
WITHSTANDING VOLTAGE	P COIL-S COIL	AC 3.75 KV r.m.s (1 MINUTE) (Mp Condition (AC 4.2KV(RMS) 2Sec)										
	P COIL -CORE	AC 3.0 KV r.m.s (1 MINUTE) (Mp Condition (AC 3.6KV(RMS) 2Sec)										
INSULATION RESISTANCE	P COIL-S COIL	100MΩ MIN DC. 500V										
LAYER SHORT	1 - 3 (4-5, 6-2 short)	AC 1.5KV Pulse 4 Time										
DC OVERLAP 90% MIN	1 -- 2	DC 5.5A										
	L(UH)	90% MIN										
QUALITY FACTOR	1 -- 2	150 MIN										
SPEC NO.		DATE			2025. 05. 26.		SHEET NO.					

SPECIFICATION

NO.	22 / 27	LLC TRANS	P / N	26OP-LM83W(EBJ30067801)
변경사항 CHANGE	(1)		(3)	
	(2)		(4)	
17.ASSIGNMENTS 1) APPLY IMPORT INSPECTION SPECIFICATION TO THE MATTERS WHICH ARE NOT GIVEN IN DRAWING SPECIFICATION METHOD. 2) PACKING METHOD FOLLOWS SPECIFICATION APPROVAL. 3) IF INDUCTANCE : REF IS NOT SATISFIED WITH GIVEN SPECIFICATION, INTERNAL STRUCTURE DISASSEMBLY INSPECTION SHOULD BE NECESSARY. 4) WIRE BUNDLE METHOD. * CENTER SOLENOID : ERROR BETWEEN THE CENTER OF OVERALL WIRE BUNDLE AREA AND THE CENTER OF WIRE BUNDLE IS REQUIRED TO BE SMALLER THAN 2MM. * PIN SOLENOID : ERROR BETWEEN THE BOTTOM BARRIER TAPE AND WINDING START POINT IS REQUIRED TO BE SMALLER THAN 2MM. * TOP SOLENOID : ERROR BETWEEN THE TOP BARRIER TAPE AND LAST WIRE IS REQUIRED TO BE SMALLER THAN 2MM. * SOLENOID AFTER : ERROR BETWEEN THE LAST WIRE OF THE WIRE BUNDLE AND START WIRE OF AFTER -SOLENOID IS REQUIRED TO BE SMALLER THAN 2MM. * TAB SOLENOID : BUNDLE SHOULD BE CONTINUED FROM PREVIOUS BUNDLE. IF BUNDLE METHOD IS 2 LAYER, BUNDLE SHOULD BE DONE FROM TOP TO BOTTOM. * SPACE SOLENOID : THE AREA BETWEEN WIRES SHOULD BE UNIFORM. ERROR BETWEEN START AND START POINT OF THE WIRE BUNDLE AREA IS REQUIRED TO BE SMALLER THAN 2MM. ERROR BETWEEN END POINT AND END POINT OF THE WIRE BUNDLE AREA IS REQUIRED TO BE SMALLER THAN 2MM. * SOLENOID : NO SPACE IS ALLOWED BETWEEN WIRES IN BUNDLING. THE SPACE BETWEEN WIRE IS REQUIRED TO BE SMALLER THAN DIAMETER OF THE WIRE*2. IN SAME LAYER, NO CROSS IS ALLOWED BETWEEN WIRES. 5) PERFORMANCE OF THIS DRAWING IS REQUIRED TO MEET PART NUMBER 26OP-LM83W 6) AUXILIARY TAPE CAN BE USED IN ORDER TO ENHANCE. BUT IT NEEDS PRIOR APPROVAL FROM DESIGN DEPARTMENT.				
SPEC NO.		DATE	2025. 05. 26.	SHEET NO.

SPECIFICATION

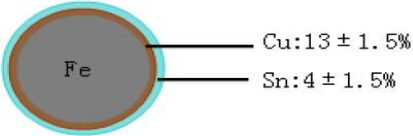
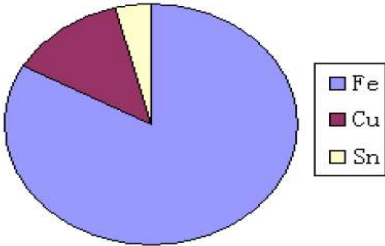
NO.	23 / 27	LLC TRANS	P / N	26OP-LM83W(EBJ30067801)
변경사항 CHANGE	(1)		(3)	
	(2)		(4)	

18.CHEMICAL ANALYSIS REPORT FOR BOBBIN

재 질 증 명 서 MATERIAL IDENTIFICATION	
원재료명 (GENERIC NAME OF MATERIAL)	LCP
원재료번호 (GRADE NO)	E4008
원재료회사 (MANUFACTURER OF MATERIAL)	SUMITOMO CHEMICAL CO LTD
난연성 (FLAMMABILITY CLASSIFICATION)	94V-0
UL 등록번호 (UL FILE NO)	E249884

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SPECIFICATION

NO. 24 / 27		LLC TRANS		P / N		26OP-LM83W(EBJ30067801)					
변경사항 CHANGE	(1)				(3)						
	(2)				(4)						
19.CHEMICAL ANALYSIS REPORT FOR CP-WIRE											
CP 线材质证明书 (MATERIALS CERTIFICATE)											
低碳钢		化学成分%									
Iron(Fe) 83% (±1.5%)	标准	C	Si	Mn	P	S					
	Q/XG173-2015	0.0026	0.002	0.05	0.009	0.006					
电解铜		化学成分%									
Copper (Cu) 13% (±1.5%)	标准	Cu		As		Sb		Bi			
	GB/T467-97	≥99.95		≤0.0015		≤0.0015		≤0.0006			
锡球		化学成分%									
Tin(Sn) 4% (±1.5%)	标准	Sn	As	Fe	Cu	Pb	Bi	Sb	Al	Cd	Zn
	GB/T728-98	99.964	0.0018	0.0027	0.0037	0.0072	0.0024	0.0140	0.0001	0.0001	0.0004
截面图					占有率%						
											
*以上的占有率是根据重量的百分比, 所以它的直径越小, 长度会越长, 但每公斤内含有的 Fe 及 Cu 成分仍不变, 因此, 此占有率同用于全规格											
We certify this paper is the certificate of our products for inspection and materials					General Manager of Quality Assurance						
					江阴六环合金线有限公司/品保部						

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SPECIFICATION

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변경 사항 CHANGE	(1)		(3)	
	(2)		(4)	

20.WIRE SPECIFICATION

도 체		도체저항 (20℃) (Ω/km) ↓	완성 외경 (mm)	실 중량 (kg/km) (Approx)	보반포함 중량 (kg) (Approx)
도체경 (mm)	가닥수				
0.10	40	67.50	0.94 ±0.15	3.40	4.4
0.10	42	64.28	0.96 ±0.15	3.64	4.7
0.10	50	54.00	1.08 ±0.15	4.22	5.3
0.10	55	49.09	1.14 ±0.15	4.96	6.0
0.10	60	45.00	1.17 ±0.15	5.20	6.2
0.10	65	41.54	1.20 ±0.15	5.66	6.7
0.10	70	38.57	1.23 ±0.15	6.20	7.2
0.10	80	33.75	1.28 ±0.15	6.89	7.9
0.10	84	32.14	1.30 ±0.15	7.27	8.3
0.10	90	30.00	1.33 ±0.15	7.78	8.8
0.10	95	28.42	1.36 ±0.15	8.30	9.3
0.10	100	27.00	1.39 ±0.15	8.52	9.5
0.10	105	25.71	1.41 ±0.15	8.95	10.0
0.10	110	24.55	1.45 ±0.15	9.38	10.4
0.10	120	22.50	1.48 ±0.15	10.23	11.2
0.10	130	20.77	1.52 ±0.15	11.08	12.1
0.10	133	20.30	1.54 ±0.15	11.18	12.2
0.10	140	19.29	1.58 ±0.15	11.77	12.8
0.10	150	18.00	1.66 ±0.15	12.12	13.1
0.10	160	16.88	1.72 ±0.15	12.93	13.9
0.10	170	15.88	1.78 ±0.15	13.74	14.7
0.10	180	15.00	1.84 ±0.15	14.55	15.5
0.10	190	14.21	1.90 ±0.15	15.36	16.4
0.10	200	13.50	1.96 ±0.15	16.17	17.2
0.10	210	12.86	2.02 ±0.15	16.98	17.9
0.10	220	12.27	2.08 ±0.15	17.79	18.8
0.10	230	11.74	2.14 ±0.15	18.59	19.6
0.04	360	38.00	1.32 ±0.15	5.60	6.6
0.05	825	13.09	2.13 ±0.15	17.85	18.9
0.08	339	12.44	3.58 ±0.15	21.55	22.6

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SPECIFICATION

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변경 사항 CHANGE	(1)		(3)	
	(2)		(4)	

21. QC Flow chart

제 / 개정	일자	개정사유	개정No.	4M QC 공정도 <TRANS>				작성	검토	승인
	07월 15일	신규 제품	QC-1					07월 15일	/	07월 15일
공정번호	공정 흐름도	공정명	작업내용	Man(사람)	Machine(장비)	Material(재료)	Method(방법)	4M		
1		원/부자재 수입검사	원자재 수입검사		Multi Meter (HP) V/C / LUPE XRF(SKYRAY)	CORE / BOBBIN / WIRE 의	수입검사 기준서 승인원			
2		권선	BOBIN에 WIRE 권선 및 PIN결선		권선기 , NIPPER 등	BOBBIN / WIRE INSULATION TAPE BARRIER TAPE 의	권선 작업지도서			
			공정검사(권,결선상태)		육안 / V/C TURN METER(대화)	권선품	작업지도서			
3		납땜 (★CTQ)	권선품 PIN 결선부위 납땜	XU TAO / XIU ZHAO YANG	DIPPING기 온도계 / TIMER	권선품 SOLDER / FLUX	온도: 460℃ MAX 시간: 선경별 납땜시간			
			공정검사(납땜상태)		육안 / LUPE	납땜 완료 권선품	SHORT, 납땜 높이			
4		조립	권선품에 CORE 조립		LCR METER (HP)	CORE / FIXING TAPE	CORE 조립 작업지 도서			
			bonding		BONDING기 (SEC-300ED / DSPM-01B) 	권선품 GAP BOND	CORE 조립 작업지 도서			
			공정검사(조립상태)		육안 / LUPE	조립 완료 제품	조립상태: GAP위치 TAPE 턴수: 3Ts			
5		1차 검사	특성(L값, 턴수)검사 외관(납땜상태 등)검사		LCR METER (HP) Turns METER(대 화) 육안/LUPE	조립 완료 제품	1차 검사 기준서			
6		합침 (★CTQ)	진공합침	WANGLIN XIAO / JIANGYANHONG	진공 합침기 TIMER / 점도계	조립 완료 제품 VARNISH	합침 작업 지도서 진공: 70~85 Hg/cm ²			
			공정검사(합침상태)		육안	합침 완료 제품	합침 상태 , 외관 PIN이음			
7		건조 (★CTQ)	건조 작업	YANG LIN / XIU ZHAO YANG	건조기 TIMER/ 온도계	합침 완료 제품	건조 작업 지도서 온도: 120 ± 5℃ 시간: 3Hr			
			공정검사(합침상태)		육안	건조 완료 제품	건조 상태 , 외관 PIN이음			
8		2차 검사	특성검사		종합계측기(TPNHUI) / 내압기/ SURGE (TOS)	건조 완료 제품	검사 기준서 승인원 특성 SPEC			
			외관 검사		육안 / LUPE PIN JIG	건조 완료 제품	검사 기준서 승인원 외관 SPEC			
9		마킹	마킹작업		자동 MARKING기	건조 완료 제품 , INK	마킹 작업 지도서 승인원 마킹 사양			
			공정검사(마킹상태)		육안	마킹 완료 제품	마킹 사양 , 위치 등 사양			
10		포장	제품 포장 진행		TAPE 절단기	완제품 , 포장 BOX 포장 TAPE	포장 작업 지도서 포장 규격			
11		출하검사	출하검사		LCR/ DCR/내압기 V/C / 육안	완제품	출하검사 기준서 출하검사 성적서			
12		출하	출하진행				출고 요청 수량 확인			

- * 작성방법
- 공정번호 : 원자재 수입검사부터 완제품 출하까지 순차적 공정 No 부여 (실제 공정 No와 일치해야함)
 - 공정흐름도 : 협력사에서 사용되는 기준에 따라 도형으로 표기
 - 공정명 : 해당공정 이름 (CTQ공정은 ★ 표기)
 - 작업내용 : 공정에 대한 작업내용
 - 사람 : CTQ공정에 한해 작업자 이름 (교대근무일경우 전/후 모두기입)
 - 장비 : 해당공정 사용장비명 및 회사명
 - 재료 : 생산 및 검사에 사용되는 원재료
 - 방법 : 생산 및 검사를 하기위한 조건(반드시 단위 및 Tolerance 기입)

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SPECIFICATION

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변경사항 CHANGE	(1)		(3)	
	(2)		(4)	

22. 4M Category

4M	4M 표준 분류 항목	세부항목	설명	1차 Vendor			2차/3차 Vendor			제출 서류
				자체 승인	LGE 승인	중요	1차 승인	LGE 승인	중요	
MAN	일반 작업/검사자 변경		외국인, 신입사원, 일용직	○			○			1. 교육결과 (자체)
	중요 작업/검사자 변경		CTO 공직 (중요공직)		○		○			1. 4M변경서
	상호 변경				○		○			2. 교육결과
MACHINE	품질책임자, 친환경 책임자 변경		품질 부서장, 친환경 책임자		○		○			3. 인수인계서(품질부서장변경시)
	생산설비 (In-Line)	교정 / 수리 Spare Part 교환		○			○			1. 4M변경서
	신규 / 수정					○			○	2. 변경전/후검사항적서
	(QA) 검사설비, Jig	교정 / 수리 Spare Part 교환		○			○			3. 설비점검자료
	신규/ 수정				○		○			4. 변경전/후실행성자료(신규,수정)
	금형	이동 신작	금형 타 업체 이동(이관) 금형 호기 추가		○			○		1. 4M변경서
	수정		성형두 수정, 교환(Punch, Die, Core)			○			○	2. 변경전/후초기검사항적서
	수리			○						3. 금형점검자료
	Line	신설,복수	사내 Line 증설		○		○			4. 변경전/후실행성자료(금형신작)
	이동		설비, Jig 변동 없음			○	○			1. 4M변경서
MATERIAL	원자재	재질/성분 변경 용성 변경 안료/도료 변경 Scrap 사용 비율 변경 색상(Color) 변경 표면 처리, 도금, 도장	1. 재질변경시 : 사전 HSMS 등록			○			○	2. 변경전/후검사항적서
	부자재	추가				○			○	3. 변경전/후실행성자료
	세척제,이형제, 보호이닐,부직포	삭제	1.업체요구에의한변경:생산성향상			○			○	4. 유해물질검사항적서
	Gasket류,Rubber류	변경	2.재질변경시:사전HSMS등록			○			○	5. 변경전/후사양승인원
	하위 부품	변경 (업체변경)	1. 재질변경시 : 사전 HSMS 등록			○			○	1. 4M변경서
	생산지	추가	1. 재질변경시 : 사전 HSMS 등록			○			○	2. 변경전/후검사항적서
	하위 Vendor	이동	1. 재질변경시 : 사전 HSMS 등록			○			○	3. 변경전/후실행성자료
	변경		1. 재질변경시 : 사전 HSMS 등록			○			○	4. 유해물질검사항적서
	검사 기준 변경	수입검사 공정검사		○			○			5. OC공정도
	공정 변경	작업방법 변경 사출/성형 조건 추가공 변경				○			○	6. 공장심사자료: 2,3차 Vendor는 1차 Vendor 실시자료
METHOD	Label/Marking		규격 Label 변경 : 규격그룹 검토		○		○			1. 4M변경서
	Packing 변경	BOX, Tray	1. 업체 요구에 의한 변경		○		○			2. 변경전/후사양승인원
	ROM 변경	ROM 버전 변경	PDP 모듈 별도 협의			○				3. 변경전/후실행성자료(Packing시행)
										4. 사양승인원
										1. 4M변경서
										2. 변경전/후검사항적서

- 1) 1차 Vendor : 제품을 조립하는 Maker
- 2) 2(3)차 Vendor : 1(2)차 Vendor에 공급하는 Maker
 - * 4M의 Vendor 정의 : 원재료를 생산 또는 조립하는 가공 업체, Agent 또는 공급권만 가지고는 있는 Supplier는 Vendor에서 제외함.
- 3) 단순 : 단순 4M으로 관련자료를 기준으로 LGE QA에서 승인함
- 4) 중요 : 관련부서 (연구소, 부품개발, 규격, 인정시험, 제품 Event)등 검토 후 승인함.

SPEC NO.		DATE	2025. 05. 26.	SHEET NO.	
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